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Medication recommendation via dual molecular modalities and multi-step enhancement

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ABSTRACT

As the integration of artificial intelligence technology with the medical field deepens, medication recommendation, as an important subfield, demonstrates immense potential value. Medication recommendation combines patient medical history with biomedical knowledge to assist doctors in determining medication combinations more accurately and safely. Existing works based on molecular knowledge neglect the 3D geometric structure of molecules and fail to learn the high-dimensional information of medications, leading to structural confusion. Additionally, it does not extract key substructures from a single patient visit, resulting in the failure to identify medication molecules suitable for the current patient visit. To address the above limitations, we propose a bimodal molecular recommendation framework named BiMoRec, which introduces 3D molecular structures to obtain atomic 3D coordinates and edge indices, overcoming the inherent lack of high-dimensional molecular information in 2D molecular structures. To retain the fast training and prediction efficiency of the recommendation system, we use bimodal graph contrastive pretraining to maximize the mutual information between the two molecular modalities, achieving the fusion of 2D and 3D molecular graphs. In addition, we propose a multi-step molecular enhancement mechanism to re-weight the molecules and optimize the utilization of the refined molecular embedding space. Our implementation on the MIMIC-III and MIMIC-IV datasets demonstrates that our method achieves state-of-the-art performance. Compared to the second-best baseline, our model improves accuracy by 0.61%, while maintaining the same level of DDI and model efficiency. Our source code is publicly available at: https://github.com/guangyunms/BiMoRec.

1. Introduction

Medication recommendation is a pivotal domain within contemporary medical informatics, aiming to provide personalized medication guidance to clinicians and patients by utilizing electronic health records (EHRs). With the rapid development of artificial intelligence (AI) in recent years, medication recommendation has witnessed numerous innovations.

Early medication recommendation approaches (An et al., 2021; Wang, Liang and Liu, 2022; Wang et al., 2017) primarily rely on the patient's most recent visit records, generating recommendations based on their current heath condition. Subsequent studies expand this approach by incorporating longitudinal historical visit records (Jin et al., 2018; Le, Tran, & Venkatesh, 2018; Zheng et al., 2021), which significantly improve recommendation accuracy. However, the complexity of treatment plans introduces a critical challenge: drug–drug interactions (DDIs),[2] which can cause adverse reactions and harm patient health. This issue is particularly pressing, as the MIMIC-III dataset reveals that patients, on average, are prescribed approximately 20 different medications (Bhoi, Lee, Hsu, Fang, & Tan, 2021).

To balance accuracy and safety, existing methods (Wu, Qiu, Jiang, Qi, & Wu, 2022; Yang, Xiao, Ma, Glass and Sun, 2021; Yang, Zeng, Wu, & Yan, 2023) model medications using both coarse-grained perspectives (medication entities) and fine-grained perspectives (molecular structures). Typically, fine-grained methods (Yang, Xiao, Ma et al., 2021; Yang et al., 2023) represent medications as 2D molecular substructures, capturing detailed relationships and modeling DDIs, thus outperforming coarse-grained methods in medication recommendation. Nevertheless, these methods still have certain limitations.

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[1] First Author and Second Author contribute equally to this work.
[2] The terms "drug" and "medication" are used interchangeably in this paper.
[3] https://pubchem.ncbi.nlm.nih.gov/.

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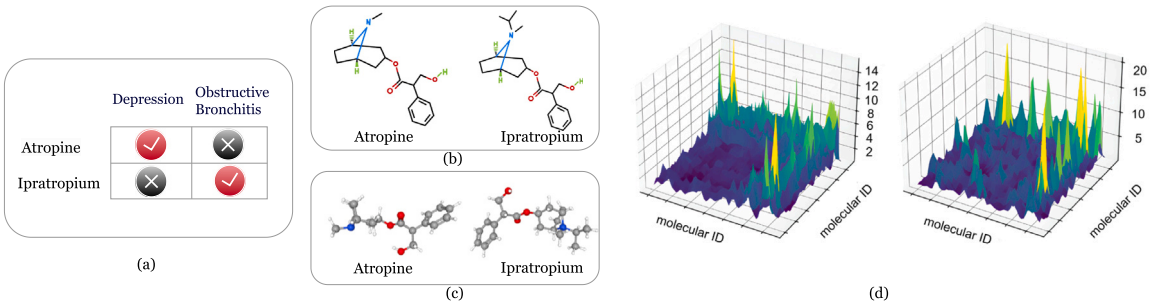


Fig. 1. The chemical characteristics of Atropine and Ipratropium. (a) Atropine for depression and Ipratropium for obstructive bronchitis; (b) 2D molecular structure representation; (c) 3D molecular structure representation; (d) Molecular similarity comparison.

Specifically, although these methods focus more finely on the 2D molecular structure of medications, they overlook the 3D spatial structure at the molecular level, leading to suboptimal performance in medication recommendations. As shown in Fig. 1(a), Atropine is used to treat depression, while Ipratropium is prescribed for obstructive bronchitis. Using the public database PubChem,[3] we illustrate the structural representations of these two medications. Fig. 1(b) displays their 2D molecular structures, which reveal a high degree of similarity, with a Tanimoto coefficient exceeding 95% according to Landrum (2013). Consequently, methods relying solely on 2D molecular structures may confuse these two medications, leading to inappropriate recommendations and raising safety concerns. However, further analysis of their 3D spatial structures, as shown in Fig. 1(c), reveals significant differences between Atropine and Ipratropium, underscoring the importance of considering 3D structural information for accurate and safe medication recommendations.

As shown in Fig. 1(d), we compute differences in the similarity matrix between molecular embeddings and fingerprints, generating a three-dimensional topographic map after sampling and nonlinear scaling. Molecular fingerprints encode structural information as bit vectors for molecular recognition. The left panel shows minimal differences in similarity compared to 2D fingerprints, with uniform and evenly distributed terrain. In contrast, the right panel, incorporating 3D modalities, reveals uneven terrain, improving distinguishability and optimizing the embedding space. This highlights that 3D representations effectively differentiate stereoisomers, such as enantiomers, which appear identical in 2D. The advantages of 3D spatial structures are as follows:

Enhanced Recommendation Accuracy. The 3D spatial structure overcomes the limitations of impaired recommendation performance caused by the similarity of 2D structures. It minimizes noise interference from 2D representations and provides a more accurate depiction of the true shape and spatial arrangement of molecules.

Alignment with Actual Medication Structures. 3D spatial structures better represent the actual state of medications, as they inherently possess three-dimensional configurations. Traditional 2D molecular structures are often fragmented into smaller substructures during modeling. However, these substructures disrupt the medication's original spatial structure, resulting in a loss of integrity and stability.

To enhance model training efficiency, this study integrates the proposed 3D concepts with 2D structures, introducing a bimodal molecular recommendation framework named **BiMoRec**. First, the 2D modality leverages a GIN network (Xu, Hu, Leskovec, & Jegelka, 2018) to extract embeddings of medication molecular structures and their substructures from molecular graphs. Next, based on the SDF file of the molecule, corresponding 3D data is retrieved, and 3D representations of the substructures are generated. A 3D geometric vector machine (Jing, Eismann, Suriana, Townshend, & Dror, 2020; Luo, Liu, & Peng, 2023) is then employed to compute high-dimensional embeddings of molecular structures and substructures. Finally, a bimodal graph contrastive learning module is developed to compare mutual information between the two modalities, facilitating joint learning of molecular representations. This framework resolves the challenges posed by the similarity of 2D molecular structures and the limitations of substructures in representing the true molecular state, while also mitigating training efficiency issues associated with the complexity of 3D structures.

Additionally, this study introduces a multi-step molecular enhancement mechanism to explore the correlation between a patient's health status and the molecular dual-granularity structure. First, within each visit, a self-attention mechanism is applied to medication molecules to strengthen their interactions. Next, relationships between molecules, diseases, medications, and procedures across multiple visits are analyzed to enhance the model's learning capacity. Finally, relevant molecules are identified based on the patient's current condition, enabling the recommendation of appropriate medications.

Specifically, the main contributions of this paper can be summarized as follows:

- We, for the first time, incorporate the 3D spatial structure information of medications to differentiate their representations from a more microscopic view, helping the model better distinguish important features from noise, reduce bias and enhance overall performance.
- We simultaneously utilize both the 2D and 3D modalities of medication molecules, along with a dual-modal graph contrastive learning module, to help the model capture key details that may be missed by a single modality. This simultaneous use provides a more comprehensive representation of molecular features, leading to a deeper understanding of molecular properties and stronger generalization capabilities for the model.
- We employed a multi-step molecular enhancement mechanism to assist the model at various levels, enabling more effective recommendations and improving both the accuracy and safety of the recommendations.
- We conduct extensive experiments on two real datasets, MIMIC-III and MIMIC-IV, and the results demonstrate that our proposed method outperforms existing methods across multiple metrics.

The structure of this paper is outlined as follows: (1) Introduction: Highlights the key innovations and the motivation for this study. (2) Related Work: Reviews significant studies and current trends in medication recommendation research. (3) Problem Definition: Defines the relevant medical concepts, entities, and the inputs and outputs used. (4) Methods: Explains the main ideas of the model and its implementation details. (5) Experiments: Describes the experimental setup, benchmarks, evaluation methods, and results. (6) Discussion: Analyzes the experimental outcomes and introduces a series of supplementary experiments. (7) Conclusion: Summarizes the key findings and suggests directions for future research.

2. Related works

In this section, we review related work in two key areas: medication recommendation and medication molecular representation.

2.1. Medication recommendation

In recent years, the field of medication recommendation has evolved through several stages, with the goal of offering personalized medication plans for patients. Early medication recommendation methods, such as instance-based approaches (Zhang, Chen, Tang, Stewart, & Sun, 2017), focused on recommending medications based on single-visit patient data. Subsequent studies introduced sequential models (Choi et al., 2016) to account for longitudinal data from patients' historical visit records.

Existing medication recommendation tasks can be divided into three categories: (1) The first approach views medication recommendation as a conventional recommendation problem, focusing on the underlying techniques in the recommendation domain. RETAIN (Choi et al., 2016), based on a two-level attention model, identifies key time points in patients' historical visits. GAMENet (Shang, Xiao, Ma, Li, & Sun, 2019) enhances medication systems by leveraging graph neural networks. COGNet (Wu et al., 2022) employs transformers for medication recommendation, adopting a translation-based approach to recommend medications from medical entities. Another study (Jin et al., 2018) employs three LSTMs to capture relationships between heterogeneous data, predicting necessary medication combinations. (2) The second approach emphasizes the relationships between medical entities, including diseases, procedures, and medications. DPR (Zheng et al., 2021) represents interactions between medical entities as a graph and designs a medication recommendation module based on graph neural networks to capture correlations among entities. DPG (Zheng et al., 2023) takes into account drug-drug interactions (DDI), constructing a medication interaction graph based on patient history and related knowledge. StratMed (Li, Liang, Hou, & Ma, 2023) employs a dual-graph architecture to balance accuracy and safety, collaboratively enhancing model performance. (3) The third approach utilizes external knowledge to strengthen relationships among medical entities, thereby improving model performance. For example, KGDNet (Mishra & Shridevi, 2024) builds personalized medical knowledge graphs by integrating EHR data with external ontologies and DDI knowledge to provide patient-specific recommendations. PROMISE (Wu, Yu, He, Gao, & Gong, 2024) enhances patient representations by leveraging multi-source external knowledge and multi-modal representation learning based on graph structures. HKRec (Zhang, Zhang, Wang, Liang, & Wang, 2024) incorporates specialized knowledge of hypertension and builds a hyper-relational knowledge graph, enabling more accurate and personalized medication recommendations.

2.2. Molecular representation learning

In existing studies that enhance medication recommendation performance using external knowledge, the majority rely on molecular knowledge of medications.

Carmen (Chen, Li, Geng, & Wang, 2023) fuses patient records with molecular representations to further differentiate molecules, enhancing recommendation outcomes. SafeDrug (Yang, Xiao, Ma et al., 2021) constructs 2D molecular modality data into graph structures and employs GNN to capture drug-drug interaction (DDI) relationships between molecules, thereby improving the safety of medication recommendations. MoleRec (Yang et al., 2023) not only utilizes 2D molecular data but also segments molecules into multiple substructures, building mappings between molecules and substructures to uncover deeper medication relationships. MEGACare (Wu, He, Mao, Li and Cambria, 2023) integrates 2D molecular data into a multi-perspective hypergraph prediction framework, leveraging molecular knowledge to enhance the representation of medical codes.

In research utilizing external knowledge, integrating molecular knowledge has become a reliable approach to improve model performance. However, previous work only considers the 2D modality of molecules, resulting in molecular representations lacking high-dimensional information. This leads to structural confusion and feature overlap among molecules, making it difficult to clearly distinguish similar molecules in the representation space, thereby reducing the overall discriminative power of the representation space.

Researchers have long recognized the importance of 3D geometric information in the molecular domain. Initially, 3D molecular structures were introduced to predict the binding sites and affinity between medications and targets (Meng, Zhang, Mezei, & Cui, 2011), solving practical challenges associated with 3D structure applications. Subsequently, studies explored the role of 3D quantitative structure–activity relationship (3D-QSAR) (Verma, Khedkar, & Coutinho, 2010) models in medication design, providing a foundation for medication prediction and optimization. KDBNet (Luo et al., 2023) utilizes 3D protein and molecular structures to predict the binding affinity between protein kinases and compounds, incorporating 3D properties often neglected by other models. Recent review articles (Isert, Atz, & Schneider, 2023) have overviewed the latest applications of 3D geometric structures in bioorganic and medicinal chemistry, highlighting their potential in structure-based medication discovery and design.

Multi-modal refers to the integration of multiple types of data or information from different sources or modalities. Molecular multimodal refers to the incorporation of various types of molecular data, such as 2D structural information and 3D geometric information, to enhance molecular representation and analysis. In the field of multi-modal molecular learning, 2D graph structures and 3D geometric structures can be viewed as different perspectives of the same molecule. Inspired by contrastive pre-training methods in the vision domain, many research efforts have begun to explore how to use 2D and 3D information jointly for molecular pre-training. For example, using two encoders to separately encode 2D and 3D molecular information while maximizing the mutual information between their representations (Stark et al., 2021). Alternatively, pre-training 2D and 3D encoders through contrastive learning and reconstruction (Liu et al., 2021). Recent studies have unified the aforementioned 2D and 3D pre-training methods and proposed a 2D graph neural network model (Zhu et al., 2022) that can be enhanced by 3D geometric features.

Our research aims to integrate molecular multi-modality without compromising the efficiency and complexity of the recommendation system, in order to address the issue of structural confusion in medication recommendation and significantly improve existing models.

3. Problem definition

3.1. Medical entity

In this study, medical entities primarily include three types: diseases, procedures, and medications, denoted as $\mathcal{D} = \{d_1, d_2, \ldots\}$, $\mathcal{P} = \{p_1, p_2, \ldots\}$, and $\mathcal{M} = \{m_1, m_2, \ldots\}$. Additionally, the molecular data come from the PubChem open-source database.

3.2. Input and output

This paper uses EHR as the data source, containing comprehensive records of patient visits and treatments. Each patient record is represented as $\mathcal{H}$, which consists of multiple visit entries $\mathcal{H} = \{v_1, v_2, \ldots, v_t\}$, where v_t corresponds to the clinical information for the t-th visit. Each visit $v_t = \{\mathcal{D}_t, \mathcal{P}_t, \mathcal{M}_t\}$ includes three elements: $\mathcal{D}_t \subset \mathcal{D}$, $\mathcal{P}_t \subset \mathcal{P}$, and $\mathcal{M}_t \subset \mathcal{M}$, representing the patient's medical data for both historical and current visits. The model's output is denoted as $\hat{\mathcal{M}}_t$, which is the predicted medication combination for the visit v_t.

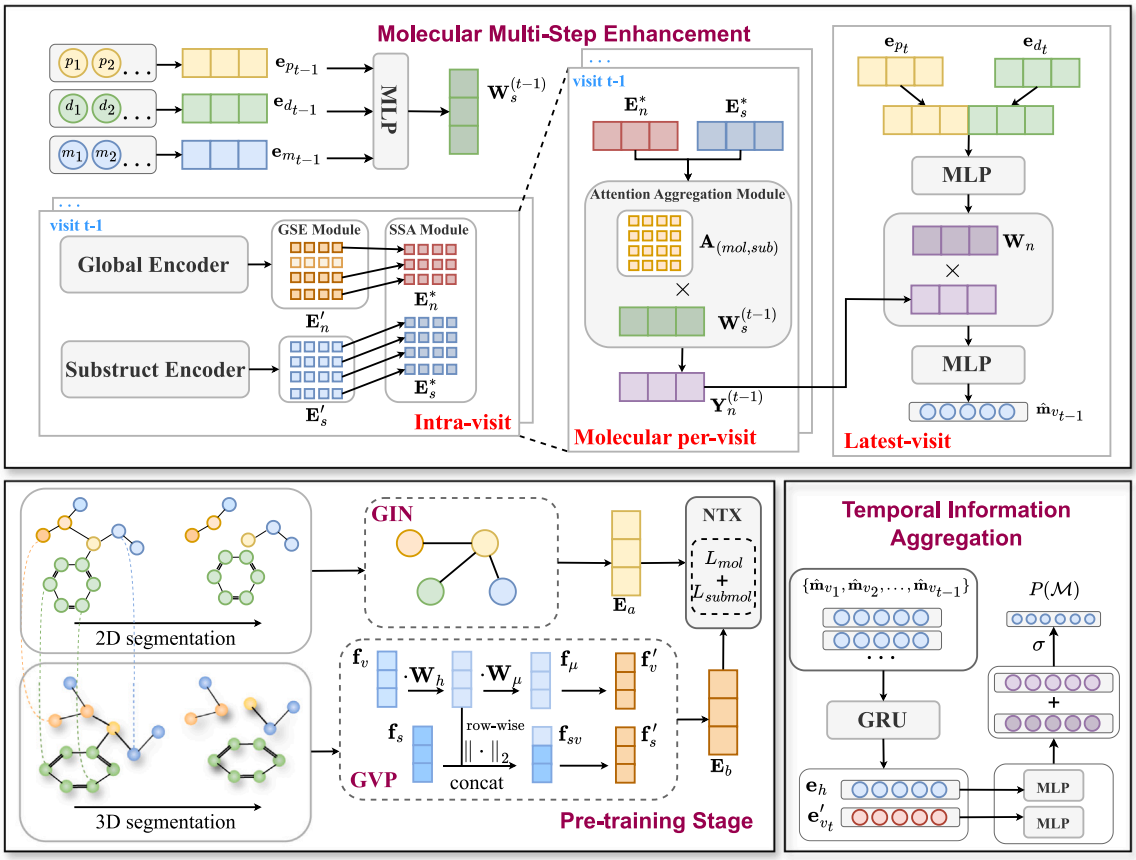


Fig. 2. The pre-training phase illustrated in the bottom left side of the figure demonstrates the bimodal fusion process of molecular structure and decomposed substructures, and its results are applied to the downstream tasks shown in the top left side. The molecular multi-step enhancement phase in the upper corner explains how molecular knowledge interacts with patient representations at the visit level. It sequentially performs molecular enhancement at three hierarchical levels: intra-visit, molecular per-visit, and latest-visit. The temporal information aggregation phase on the lower right side of the figure models the sequence relationships of the interaction results and maps them to recommendation probabilities, obtaining the final recommendation results.

3.3. DDI matrix

DDIs are a critical concern in medication recommendations, as they highlight combinations of medications that may present confirmed safety risks to patients. To enhance the safety of medication therapy, it is essential to exclude combinations that carry such risks, thereby minimizing the occurrence of adverse events.

We obtain our DDI data from the Adverse Event Reporting Systems (Hoffman, Dimbil, Erdman, Tatonetti, & Overstreet, 2014). This information is represented in a binary matrix $\mathbf{M}^{ddi} \in \{0,1\}^{|\mathcal{M}|\times|\mathcal{M}|}$, where an entry of $\mathbf{M}^{ddi}_{ij} = 1$ denotes an interaction between medications m_i and m_j. A high occurrence of DDIs in this matrix indicates potential safety concerns in the recommended medication combinations. For example, in the MIMIC-III dataset (Johnson et al., 2016), the actual rate of DDIs is approximately 8% (Yang, Xiao, Ma et al., 2021). This indicates that a considerable number of medication combinations may pose safety risks to patients, emphasizing that drug-drug interactions are a significant concern in real-world clinical settings. Special attention is needed to avoid recommending combinations that could lead to adverse reactions.

4. Methods

Fig. 2 illustrates our model, consisting of three primary components.

(1) In the pre-training stage, we integrate dual-modality molecular information using contrastive learning. This allows us to obtain a molecular encoding module and a substructure encoding module that incorporate knowledge from both modalities. It can also be understood as the 3D modality teacher model transferring knowledge into the 2D model.

(2) In the molecular multi-step enhancement phase, our goal is to encode the molecules and their substructures using the encoder developed in the pre-training stage, generating embedding matrices. Through a multi-step enhancement mechanism, we apply triple enhancement of molecular representations at the visit level. This process increases the weight of key molecules to identify critical medications.

(3) In the temporal information aggregation stage, our objective is to integrate patient progression information derived from medication representations at different temporal points. By adjusting the probability bias of each medication based on the patient's current health condition, we can recommend medications that exceed a predefined threshold.

4.1. Pre-training

In this phase, our aim is to integrate the dual-modality information of molecules, which involves the representation of 2D and 3D structures as well as mutual information learning. To ensure the training efficiency of the model and reduce complexity, a contrastive learning pre-training method is employed.

For the 2D molecular modality, we first represent it as a graph structure. The medication molecule consists of atoms and bonds, where atoms serve as nodes and chemical bonds as edges. Given a molecule G, the representation of the entire graph is generated through l layers of Graph Neural Networks (GNN).

In this paper, we use GIN (Xu et al., 2018), a variant of GNN, as the encoder for medication molecular graphs to obtain holistic-level representations, which is commonly used in recent work related to medication recommendation and molecular research (Jiang, Feng, Chen, Li, & He, 2021; Miao, Liu, & Li, 2022; Wang et al., 2021). GIN iteratively updates node features by aggregating features of neighboring nodes, i.e., updating representations of atomic nodes through aggregation strategies. The node features used as input for the GIN in the model include the chemical properties and structural characteristics of atoms.[4] These features capture the fundamental structural and electronic properties of atoms within the molecule. We use the RDKit library[5] to extract these features. In general, we have:

$$\mathbf{f}_a^{(l)} = \text{UPDATE}\left(\mathbf{f}_a^{(l-1)}, \mathbf{f}_b^{(l-1)}, \mathbf{e}_{(a,b)} | b \in \mathcal{N}(a)\right), \tag{1}$$

$$\mathbf{E}_g = \text{READOUT}\left(\left\{\mathbf{f}_a^{(l)} | a \in V\right\}\right), \tag{2}$$

where $\mathbf{f}_a^{(l)}$ is the node feature of node a in the layer l, $\mathbf{e}_{(a,b)}$ is the edge feature of edge (a, b), and $\mathcal{N}(a)$ denotes the neighbor of node a on the graph. UPDATE function updates the feature representation of a node by aggregating its own features and neighboring features via weighted summation. Subsequently, READOUT function is employed to aggregate all the node features from the lth layer.

Through the aforementioned process, the representations of all medication molecules are obtained, forming an embedding matrix $\mathbf{E}_n \in \mathcal{R}^{|\mathcal{M}| \times dim}$, where each row represents a medication molecule. This matrix will be used in the subsequent contrastive learning phase.

For the 3D molecular modality, we represent the molecular structure as a graph using the 3D coordinates of atoms, with atoms serving as nodes. An edge is defined between two atoms if their distance is less than 4.5 Å (Townshend et al., 2020). In the constructed 3D molecular graph, each node and edge is associated with geometric features such as 3D coordinates and scalar properties that describe chemical characteristics. To encode this structure, we utilize Geometric Vector Perceptrons (GVPs) (Jing et al., 2020), which are specifically designed to handle 3D data. GVPs directly learn structural representations from raw atomic coordinates without requiring features invariant to rotation and translation.

In contrast, Equivariant Graph Neural Network (EGNN) models and related equivariant methods (Han, Rong, Xu, & Huang, 2022; Satorras, Hoogeboom, & Welling, 2021) involve higher algorithmic complexity. This arises from the need to construct features based on relative distances and perform fully connected message passing when updating node features. Moreover, EGNN models capture geometric information through scalar features, but the lack of explicit incorporation of molecular vector features limits their ability to capture directional information.

Similarly, 3D graph contrastive learning methods (Li, Zhou, Xu, Dou, & Xiong, 2022; Wang, Wang, Cao and Barati Farimani, 2022) rely on data augmentation techniques such as rotation and translation to generate positive and negative sample pairs for learning consistent molecular representations from various perspectives. However, our approach bypasses the need for data augmentation, directly modeling geometric and scalar features from raw data.

Compared to these methods, GVP excels by directly extracting vector features of 3D molecules from raw coordinates without augmentation, achieving superior performance and computational efficiency in tasks like molecular representation (Luo et al., 2023; Wu, Wu, Radev, Xu and Li, 2023; Yu et al., 2024).

In actual model construction, we use the tuple $(\mathbf{f}_s, \mathbf{f}_v)$ to represent the scalar features and vector features of atom independently. The molecular GVP transforms node and edge features through graph convolution layers to obtain the representation of the input molecule. The GVP model uses atomic symbol and implicit valence as scalar features. Additionally, GVP incorporates the 3D coordinates of atoms as vector features, which are normalized to better capture the geometric information of the molecular structure, enhancing its ability to understand both topological and spatial relationships within the molecule. The features are extracted using the same RDKit library as that used for GIN. The specific process of feature learning and transformation is as follows:

$$\mathbf{f}_{sv} = \text{CONCAT}(\|\mathbf{W}_h \mathbf{f}_v\|_{2,\text{row-wise}}, \mathbf{f}_s), \tag{3}$$

$$\mathbf{f}'_s = \sigma(\mathbf{W}_v \mathbf{f}_{sv} + b), \tag{4}$$

$$\mathbf{f}_\mu = \mathbf{W}_\mu \mathbf{W}_h \mathbf{f}_v, \tag{5}$$

$$\mathbf{f}'_v = \sigma(\|\mathbf{f}_\mu\|_{2,\text{row-wise}}) \odot \mathbf{f}_\mu, \tag{6}$$

where W_h, W_v, and W_μ are three weight matrices for transformation, b is a bias term, $\odot$ represents element-wise multiplication, σ is a nonlinear function, and CONCAT concatenates the row-wise L2 norm with $\mathbf{f}_s$. After passing through the GVP graph convolution layer, $(\mathbf{f}_s, \mathbf{f}_v)$ is transformed into $(\mathbf{f}'_s, \mathbf{f}'_v)$. Each node aggregates embeddings from its neighboring nodes and edges to update its own representation.

After the final layer of molecular GVP, a global additive pooling operation is applied to aggregate all node representations into the scalar representation of the input medication, obtaining the 3D embedding matrix $\mathbf{E}_{nt} \in \mathcal{R}^{|\mathcal{M}| \times dim}$ of the molecule, where each row represents a molecule's representation.

For molecular substructures, existing work demonstrates that the properties of medication molecules are partially determined by specific substructures, and the treatment of a patient's disease also depends on the functionality of certain substructures. Our goal is to reconstruct the 3D structures of substructures to bring them closer to the actual molecular structures. We first use the BRICS (Yang et al., 2023) method to decompose medication molecules into substructures, and then complete the 3D structure of the substructures using the API provided by RDKit (Landrum, 2013). We decompose all medication molecules to obtain a substructure set S, and finally obtain the embedding matrices of the two different modalities of the substructures $\mathbf{E}_s, \mathbf{E}_{st} \in \mathcal{R}^{|S| \times dim}$, where each row represents the embedding of a specific substructure.

In the subsequent process, contrastive learning achieves two objectives. The first objective is to make molecules with the same index i more similar. The second objective is to enforce differences between negative samples $\mathbf{E}_{a_i}$ and $\mathbf{E}_{b_k}$, where $i \neq k$, i and k represent the i-th embedding in the 2D embedding table and the k-th embedding in the 3D embedding table. We employ the commonly used NTXent loss (Chen, Kornblith, Norouzi, & Hinton, 2020; Luo et al., 2023) to jointly optimize our model:

$$\text{L}(\mathbf{E}_\text{a}, \mathbf{E}_\text{b}) = -\frac{1}{N} \sum_{i=1}^{N} \left[\log \frac{e^{sim(\mathbf{E}_{a_i}, \mathbf{E}_{b_i})/\tau}}{\sum_{k \neq i}^{N} e^{sim(\mathbf{E}_{a_i}, \mathbf{E}_{b_k})/\tau}} \right], \tag{7}$$

$$sim(\mathbf{E}_a, \mathbf{E}_b) = \mathbf{E}_a \cdot \mathbf{E}_b / (\|\mathbf{E}_a\| \|\mathbf{E}_b\|), \tag{8}$$

where $sim(\cdot)$ is the cosine similarity and τ is a temperature parameter which acts as a weight for the most similar negative pair.

We apply this loss function to molecules and their substructures by inputting $\mathbf{E}_n$ and $\mathbf{E}_{nt}$ into the aforementioned formula to obtain $\mathcal{L}_{mol}$, and inputting $\mathbf{E}_s$ and $\mathbf{E}_{st}$ into the same formula to obtain $\mathcal{L}_{submol}$. Finally, we combine them to obtain the total loss function.

$$\mathcal{L}_{mol} = \text{L}(\mathbf{E}_n, \mathbf{E}_{nt}), \tag{9}$$

$$\mathcal{L}_{submol} = \text{L}(\mathbf{E}_s, \mathbf{E}_{st}), \tag{10}$$

$$\mathcal{L}_{total} = \mathcal{L}_{mol} + \mathcal{L}_{submol}. \tag{11}$$

[4] Chemical properties include atomic number, formal charge, hybridization state, and aromaticity; structural characteristics include chirality tag, degree, number of hydrogen atoms, number of radical electrons, and whether the atom is part of a ring.

[5] https://www.rdkit.org/.

4.2. Multi-step molecular enhancement

At this stage, we will enhance the medication molecules at three levels: intra-visit, molecular per-visit, and latest-visit, to obtain the key molecular representations required by the patients.

We encode the diseases, procedures, and medications as part of the patient's current health environment. We first extract the sets of diseases, procedures, and medications at each time point from the visit v_{t-1}. We define three learnable embedding matrices, $\mathbf{E}_d$, $\mathbf{E}_p$, and $\mathbf{E}_m$, corresponding to diseases, procedures, and medications. We generate $\mathbf{e}_{d_i}$, $\mathbf{e}_{p_i}$, and $\mathbf{e}_{m_i}$ by mapping diseases d_i, procedures p_i, and medications m_i to the embedding space, and use them as entity-level representations shared globally for downstream tasks. We obtain the set representations $\mathbf{e}_{d_{t-1}}$, $\mathbf{e}_{p_{t-1}}$, and $\mathbf{e}_{m_{t-1}}$ by mapping all diseases, procedures, and medications through their corresponding embedding matrices.

$$\mathbf{e}_{d_{t-1}} = d_{t-1}\mathbf{E}_d, \quad \mathbf{e}_{p_{t-1}} = p_{t-1}\mathbf{E}_p, \quad \mathbf{e}_{m_{t-1}} = m_{t-1}\mathbf{E}_m. \tag{12}$$

Finally, we concatenate the three vectors of equal length, resulting in a single vector with three times the original length, forming the visit-level representation for the visit at time $t-1$.

$$\mathbf{e}'_{v_{t-1}} = \mathrm{CONCAT}(\mathbf{e}_{d_{t-1}} \| \mathbf{e}_{p_{t-1}} \| \mathbf{e}_{m_{t-1}}). \tag{13}$$

Then, a multi-layer perceptron (MLP) converts the dimension of the vector $\mathbf{e}'_{v_{t-1}}$ to the total number of molecular substructures, and a sigmoid function modulates the relevance of each substructure at time $t-1$.

$$\mathbf{W}_s^{(t-1)} = \sigma(\mathrm{MLP}(\mathbf{e}'_{v_{t-1}})). \tag{14}$$

Following the initialization process outlined above, we will now provide a detailed description of our molecular multi-enhancement mechanism.

4.2.1. Intra-visit enhancement

After completing the pre-training process in the previous stage, two encoders that integrate dual molecular modalities were derived: a global encoder and a substructure encoder, for use in subsequent molecular representation tasks.

We employ the global encoder and substructure encoder trained during the pretraining phase to encode molecular structures and their substructures, obtaining $\mathbf{E}'_n$ and $\mathbf{E}'_s$. For substructures, as the function of molecules is realized through a set of substructures, it is necessary to model the interactions between these substructures. Therefore, we introduce a Substructure Self-Attention (SSA) module (Lee et al., 2019) to achieve this goal. We use the embedding matrix $\mathbf{E}'_s$ of all substructures as input to establish an attention mechanism between each substructure, and finally output an embedding matrix of equal size $\mathbf{E}^*_s \in \mathbb{R}^{|S| \times dim}$. The self-attention layer is defined as follows:

$$\mathrm{H} = \mathrm{LayerNorm}(\mathbf{E}'_s + \mathbf{A}_{\mathrm{self}}(\mathbf{E}'_s)), \tag{15}$$

$$\mathbf{E}^*_s = \mathrm{SSA}(\mathbf{E}'_s) = \mathrm{LayerNorm}(\mathrm{H} + \mathrm{MLP}(\mathrm{H})). \tag{16}$$

$\mathcal{A}_{self}$ denotes the self-attention mechanism (Vaswani et al., 2017) defined by the following formula:

$$\mathbf{A}_{self}(\mathrm{X}) = \mathrm{Softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d}}\right)\mathbf{V}, \tag{17}$$

where d represents the dimensionality of the key embeddings, and $\mathbf{Q}$, $\mathbf{K}$, and $\mathbf{V}$ are the query, key, and value matrices derived from the input $\mathbf{X}$.

For molecular structures, we need to select from the embedding table $\mathbf{E}'_n$ based on the patient's visit data and perform molecular enhancement. We input $\mathbf{E}'_n$ into the global selection enhancement module (GSE) to obtain the enhanced result $\mathbf{E}^*_n \in \mathbb{R}^{|\mathcal{M}| \times dim}$:

$$\mathbf{E}^*_n = \mathrm{GSE}(\mathrm{SSA}(\mathbf{E}'_n)). \tag{18}$$

Specifically, the GSE module searches for the corresponding embedding vectors based on the medication molecule combinations relevant to the current context, constructs a small embedding matrix, and feeds it into the attention module. This process generates an enhanced matrix, which is then used to update the global molecular embedding matrix.

4.2.2. Molecular per-visit enhancement

At this stage, we will perform molecular enhancement based on each visit. Specifically, we input the molecular-level representation and substructure interaction representation, $\mathbf{E}^*_n$ and $\mathbf{E}^*_s$, into the Attention Aggregation Module to perform attention-based aggregation. When calculating the attention coefficients, we use a mask matrix to adjust the attention matrix.

$$\mathbf{A}_{(mol,sub)}(\mathrm{X}, \mathrm{Y}) = \mathrm{Softmax}\left(\mathrm{mask}\left(\frac{\mathbf{Q}_x\mathbf{K}_y^\top}{\sqrt{d_{max}}}\right)\right), \tag{19}$$

where $d_{\max}$ is the maximum embedding dimension of $\mathbf{X}$ and $\mathbf{Y}$, and the mask function replaces the set of substructures that lack corresponding mappings with the minimum value.

$$\forall s_i \in S, \mathbf{y}_{m_i} = \mathbf{A}_{(mol,sub)}\mathbf{W}_s^{(t-1)}\mathbf{E}^*_{s_i}, \tag{20}$$

where $\mathbf{E}^*_{s_i}$ is the vector of the corresponding substructure s_i in $\mathbf{E}^*_s$. Then, we obtain the contextual representation of the substructure at the current time sequence through the attention matrix.

Ultimately, we derive all the aggregated molecular embedding representations $\mathbf{Y}_n^{(t-1)}$ for the visit at time $t-1$.

4.2.3. Latest-visit enhancement

To relate the aggregated molecular embedding to the patient's latest health condition, we use the most recent patient visit as a reference for final enhanced. Through an MLP layer, we obtain the weights of the latest patient representation for the elements of the molecular set to calibrate the molecular embedding accordingly.

$$\mathbf{e}_{d_t} = d_t\mathbf{E}_d, \quad \mathbf{e}_{p_t} = p_t\mathbf{E}_p, \tag{21}$$

$$\mathbf{e}'_{v_t} = \mathrm{CONCAT}(\mathbf{e}_{d_t} \| \mathbf{e}_{p_t}), \tag{22}$$

$$\mathbf{W}_n = \sigma(\mathrm{MLP}(\mathbf{e}'_{v_t})). \tag{23}$$

$\mathbf{e}_{d_t}$ and $\mathbf{e}_{p_t}$ are the embeddings of the diagnosis and procedure representations from the patient's current visit. By concatenating these embeddings and passing them through an MLP layer, we obtain the weight vector $\mathbf{W}_n$. Finally, we correct the molecular embedding matrix by multiplying it with a diagonal matrix, and then adjust the dimensions through an MLP network to obtain the representation $\hat{\mathbf{m}}$ for each medication.

$$\hat{\mathbf{m}}_{v_{t-1}} = \mathrm{MLP}(\mathbf{W}_n \mathrm{Y}_n^{(t-1)}). \tag{24}$$

By enhancing the molecular representations based on recent patient visits, the most recent patient conditions were incorporated into the molecular matrix for each visit.

4.3. Temporal information aggregation

The purpose of this stage is to integrate the medication representations generated for the patient at each time point. This representation is derived from multiple dynamic enhancements of molecules, ultimately yielding the recommendation probabilities for all medications.

By embedding the multiple medication representations generated in the previous stage $\{\hat{\mathbf{m}}_{v_1}, \hat{\mathbf{m}}_{v_2}, \ldots, \hat{\mathbf{m}}_{v_{t-1}}\}$ and inputting them into a gated recurrent unit GRU(·) to model the sequential relationships, $\{v_1, v_2, \ldots, v_{t-1}\}$ are patient visits $t-1$ times, we then process the outputs through a multilayer perceptron. Finally, we concatenate the results with the patient's most recent representation $\mathbf{e}'_{v_t}$ to generate the final medication probabilities.

$$\mathbf{e}_h = \mathrm{GRU}(\hat{\mathbf{m}}_{v_1}, \hat{\mathbf{m}}_{v_2}, \ldots, \hat{\mathbf{m}}_{v_{t-1}}), \tag{25}$$

$$P(\mathcal{M}) = \sigma(\mathrm{MLP}(\mathbf{e}_h) + \mathrm{MLP}(\mathbf{e}'_{v_t})), \tag{26}$$

Using the nonlinear activation function σ, the input is converted into the predicted probability $P(\mathcal{M})$ for each medication. If the probability exceeds a certain threshold, the recommendation for this medication is accepted; otherwise, the recommendation is rejected. Ultimately, this process yields the predicted medication combination $\hat{\mathcal{M}}_t$ for the current visit v_t.

4.4. Model training

Medication recommendation can be framed as a multi-label binary classification task. Thus, we use the binary cross-entropy loss function L_{bce} and the multi-label margin loss function L_{multi}. Additionally, given the unique nature of medication recommendation, we use the medication DDI loss function L_{ddi} (Yang et al., 2023), which calculates the probability of DDI occurrence by identifying medication pairs with potential DDI risk in the medication combination. The definitions of the three loss functions can be defined as:

$$\mathcal{L}_{bce} = -\sum_{i=1}^{|\mathcal{M}|} m_i \log(\hat{m}_i) + (1 - m_i)\log(1 - \hat{m}_i), \tag{27}$$

$$\mathcal{L}_{multi} = \sum_{i,j:m_i=1,m_j=0} \frac{\max(0, 1 - (\hat{m}_i - \hat{m}_j))}{|\mathcal{M}|}, \tag{28}$$

$$\mathcal{L}_{ddi} = \sum_{i=1}^{|\mathcal{M}|}\sum_{j=1}^{|\mathcal{M}|} \mathbf{M}_{ij}^{ddi} \cdot \hat{m}_i \cdot \hat{m}_j, \tag{29}$$

where m_i represents the true value of the i-th medication during the current visit, while $\hat{m}_i$ represents the model's predicted value for the ith medication, which are both binary variables.

Although DDI reflects the potential risk of medication combinations, from a clinical perspective, certain adverse reactions are acceptable. Pursuing the lowest DDI rate without considering efficacy may not result in the optimal medication combination. Therefore, we must set a threshold for DDI; if this threshold is surpassed, the weight assigned to DDI increases. When using the composite loss function, we adopt the same research methodology as before (Yang et al., 2023) to achieve a balance between the accuracy and safety of the medication combinations.

$$\mathcal{L} = \alpha(\beta\mathcal{L}_{bce} + (1 - \beta)\mathcal{L}_{multi}) + (1 - \alpha)\mathcal{L}_{ddi}, \tag{30}$$

$$\alpha = \begin{cases} 1 & \rho < \phi \\ \min\{1, \exp(\tau(1 - \frac{\rho}{\phi}))\} & \rho \geq \phi, \end{cases} \tag{31}$$

where β is hyperparameters, and the controllable factor α is relative to DDI rate, $\phi \in (0, 1)$ is a DDI acceptance rate, ρ is the current DDI rate and τ is a hyper-parameter.

5. Experiments

This section provides an overview of the datasets used and a comparison of the results from the baseline models and our model. Additionally, we present an explanation of the evaluation metrics and the specific experimental setup, including model configuration and parameter selection.

5.1. Datasets

We utilize the MIMIC-III (Johnson et al., 2016) and MIMIC-IV (Johnson et al., 2023) datasets, which originate from clinical data in the intensive care unit (ICU). These datasets include clinical records, physiological measurements, laboratory monitoring data, and medication records of ICU patients. We follow the same data processing methods as previous studies. The detailed statistics of the processed datasets are presented in the Table 1.

Molecular SDF files are sourced from the PubChem database (Kim et al., 2021), and the 3D data for substructures is generated using functions from the RDKit library.

Table 1
Statistics of the datasets.

Item	MIMIC-III	MIMIC-IV
# patients	6350	60,125
# visits events	15,032	156,810
# diseases	1958	2000
# procedures	1430	1500
# medications	131	131
avg. # of visits	2.37	2.61
avg. # of medications	11.44	6.66

5.2. Baselines

To verify the performance of our model, we chose the following high-performance methods as the baseline model for comparison.

LR (Logistic Regression) is a widely employed linear classification algorithm that calculates the probability of an instance belonging to a specific category based on a weighted combination of input features. It is commonly used for probability prediction and both binary and multi-class classification tasks.

ECC (Read, Pfahringer, Holmes, & Frank, 2011) (Ensemble of Classifier Chains) utilizes a series of interconnected classifiers to improve prediction accuracy, where the output of one classifier serves as the input for the next. This method is well-suited to multi-label classification tasks and can effectively enhance the model's overall performance.

RETAIN (Choi et al., 2016) is an attention-based model that analyzes patient time series for precise disease prediction and management. It provides medication recommendations by adaptively identifying critical clinical events in a patient's medical history.

LEAP (Zhang et al., 2017) optimizes medication efficacy by modeling label dependencies and decomposes the treatment process into a sequential procedure.

GAMENet (Shang et al., 2019) is a medication recommendation model that integrates the strengths of graph neural networks with memory networks and effectively discerns patterns and temporal sequences within medical data, thus improving the accuracy of its predictions.

SafeDrug (Yang, Xiao, Ma et al., 2021) employs the combination of patients' health conditions and medication-related molecular knowledge. This approach, by reducing the impact of DDIs, offers safer medication combinations.

MICRON (Yang, Xiao, Glass and Sun, 2021) recommends medications based on the dynamic changes in a patient's health condition. It updates medication combinations according to the patient's new symptoms to improve therapeutic outcomes while reducing potential side effects.

COGNet (Wu et al., 2022) utilizes the Transformer architecture for medication recommendations, using a translation approach to derive medications from illnesses. It also features a copy mechanism to integrate beneficial medications from past prescriptions into new recommendations.

MoleRec (Yang et al., 2023) investigates the importance of specific molecular substructures in medications. This approach enhances the precision of medication recommendations by utilizing finer molecular representations.

CausalMed (Li et al., 2024) identifies causal relationships between medical entities through causal discovery, accounting for dynamic differences under varying health conditions and translating these into causally linked medication recommendations.

EGNN (Satorras et al., 2021) preserves equivariance by updating both scalar and vector features through message passing. By integrating coordinate updates, it effectively captures molecular structures, making it well-suited for dynamic geometric modeling of molecules.

SE(3)-Transformer (Fuchs, Worrall, Fischer, & Welling, 2020) explicitly integrates vector representations into a direction- and rotation-equivariant model while extending the self-attention mechanism to an SE(3)-equivariant architecture, enabling more effective encoding of directional and rotational information.

Table 2
The performance of each model on the test set regarding accuracy and safety. The best and the runner-up results are highlighted in bold and underlined respectively.

Model	MIMIC-III					MIMIC-IV				
	Jaccard↑	DDI↓	F1↑	PRAUC↑	Avg.#Med	Jaccard↑	DDI↓	F1↑	PRAUC↑	Avg.#Med
LR	0.4924	0.0830	0.6490	0.7548	16.0489	0.4569	0.0783	0.6064	0.6613	8.5746
ECC	0.4856	0.0817	0.6438	0.7590	16.2578	0.4327	0.0764	0.6129	0.6530	8.7934
RETAIN	0.4871	0.0879	0.6473	0.7600	19.4222	0.4234	0.0936	0.5785	0.6801	10.9576
LEAP	0.4526	0.0762	0.6147	0.6555	18.6240	0.4254	0.0688	0.5794	0.6059	11.3606
GAMENet	0.4994	0.0890	0.6560	0.7656	27.7703	0.4565	0.0898	0.6103	0.6829	18.5895
SafeDrug	0.5154	**0.0655**	0.6722	0.7627	19.4111	0.4487	**0.0604**	0.6014	0.6948	13.6943
MICRON	0.5219	0.0727	0.6761	0.7489	19.2505	0.4640	0.0691	0.6167	0.6919	12.7701
COGNet	0.5312	0.0839	0.6744	0.7708	27.6335	0.4775	0.0911	0.6233	0.6524	18.7235
MoleRec	0.5293	0.0726	0.6834	0.7746	22.0125	0.4744	0.0722	0.6262	0.7124	13.4806
CausalMed	0.5389	0.0709	0.6916	0.7826	20.5419	0.4899	0.0677	0.6412	0.7338	14.4295
BiMoRec	**0.5422**	0.0707	**0.6944**	**0.7858**	20.3989	**0.4917**	0.0696	**0.6425**	**0.7366**	13.5291

5.3. Evaluation metrics

We employ four primary metrics, Jaccard, DDI rate, F1, and PRAUC, to assess the effectiveness of our approach. The role of each metric in our study is outlined below.

Jaccard (Jaccard similarity score) quantifies the similarity between two sets by dividing the size of their intersection by the size of their union. In medication recommendation, a higher Jaccard score indicates increased alignment between the predicted and actual medication combinations, indicating improved accuracy.

DDI (Drug-Drug Interaction rate) evaluates the incidence of adverse reactions in the recommended combinations. A lower rate indicates greater safety in the drug combinations.

F1 (F1-score) combines precision and recall, which reflects the model's capacity to accurately detect correct medications while ensuring comprehensive coverage.

PRAUC (Precision–Recall Area Under Curve) evaluates model performance across different recall levels, demonstrating the capability to maintain precision with increasing recall.

Avg.# of Medications (Average number of Medications) quantifies the average number of medications in each recommendation result. A higher value suggests that each recommendation includes more medications, which may increase the complexity of clinical treatment and the risk of adverse reactions. Conversely, a lower value implies that the medication combinations may be safer and reduce unnecessary medication use. Notably, this metric serves as a reference only, and the size of the combination should not be used as a strict evaluation standard.

5.4. Setup protocol

In this subsection, we offer a comprehensive overview of the experimental environment setup, parameter selection, model configuration, and sampling methods for the testing phase.

5.4.1. Experimental environment

The experiments were conducted on an Ubuntu 22.04 system equipped with 30 GB of memory, 12 CPUs, and a 24 GB NVIDIA RTX3090 GPU, using PyTorch 2.0.0 and CUDA 11.7.

5.4.2. Configuration and parameter

For the embedding matrices of entities E_d, E_p, and E_m, we set 128 as the embedding size, initialized in the range of −0.1 to 0.1. For the molecular GNN, we implement a 4-layer GIN with a hidden size of 128. For each molecular graph, the initial set of 9-dimensional node features comprise atom count, chirality, and other additional atomic features. The 3-dimensional edge features include bond type, bond stereochemistry, and conjugated bonds. The molecular 3D features include molecular conformations, 3D coordinates, RBF distances, and normalized interatomic vectors. Additionally, we implement a 3-layer GVP, where the hidden dimensions for node feature scalars and vectors are 128 and 64, respectively, and the hidden dimensions for edge feature scalars and vectors are 32 and 1, respectively. For the GRU, we configure the same number of hidden units as the number of medications. For the loss functions, we define the thresholds $\delta = 0.5$, $\tau = 2.5$, $\beta = 0.95$, and $\phi = 0.06$. We conduct training for 300 epochs during the pretraining stage and 20 epochs during the training stage.

Table 3
Comparison of training and inference times for different methods.

Method	Convergence epoch	Training time /Epoch (s)	Total training time (s)	Inference time (s)
GAMENet	39	45.31	1767.09	19.27
SafeDrug	54	38.32	2069.28	20.15
MICRON	40	17.48	699.20	14.48
COGNet	103	38.85	4001.55	142.91
MoleRec	25	249.32	6233.00	32.10
CausalMed	33	164.77	5437.41	18.29
Pre-training	300	0.71	214.02	–
BiMoRec	15	416.61	6249.15	15.16

5.4.3. Sampling approach

Given the limited availability of publicly accessible EHR data, we utilize bootstrapping techniques in this phase, following the methodology outlined in Yang, Xiao, Ma et al. (2021). As discussed in Dauji and Bhargava (2016) and Picheny, Kim, and Haftka (2010), this technique is particularly effective for small sample size scenarios.

5.5. Performance comparison

In this section, we compare our model with baseline models, assessing model accuracy and safety using key metrics. For baseline methods with a provided test model, we utilize the baseline model directly. For methods without a provided test model, we retrain the models and perform evaluations. We use the optimal parameter settings described in the respective papers to derive results from each baseline method. Table 2 presents the comparative results.

To provide a clearer comparison between the baseline and our model, we plot the evaluation metrics separately, as shown in Fig. 3. Jaccard, F1, and PRAUC evaluate the model's accuracy, while DDI assesses its safety.

We also evaluate the operational efficiency of the models, including the number of convergence epochs, training time per epoch, total training time, and inference time, along with the time efficiency during the pre-training phase, as shown in Table 3.

Among the various baseline methods, LR and ECC, which employ traditional recommendation techniques, demonstrate low accuracy. LEAP decomposes treatment recommendations into a sequential decision process, leveraging sequential relationships to enhance effectiveness, but it does not surpass traditional methods. RETAIN also introduces a sequence model but leads to an excessively high DDI rate.

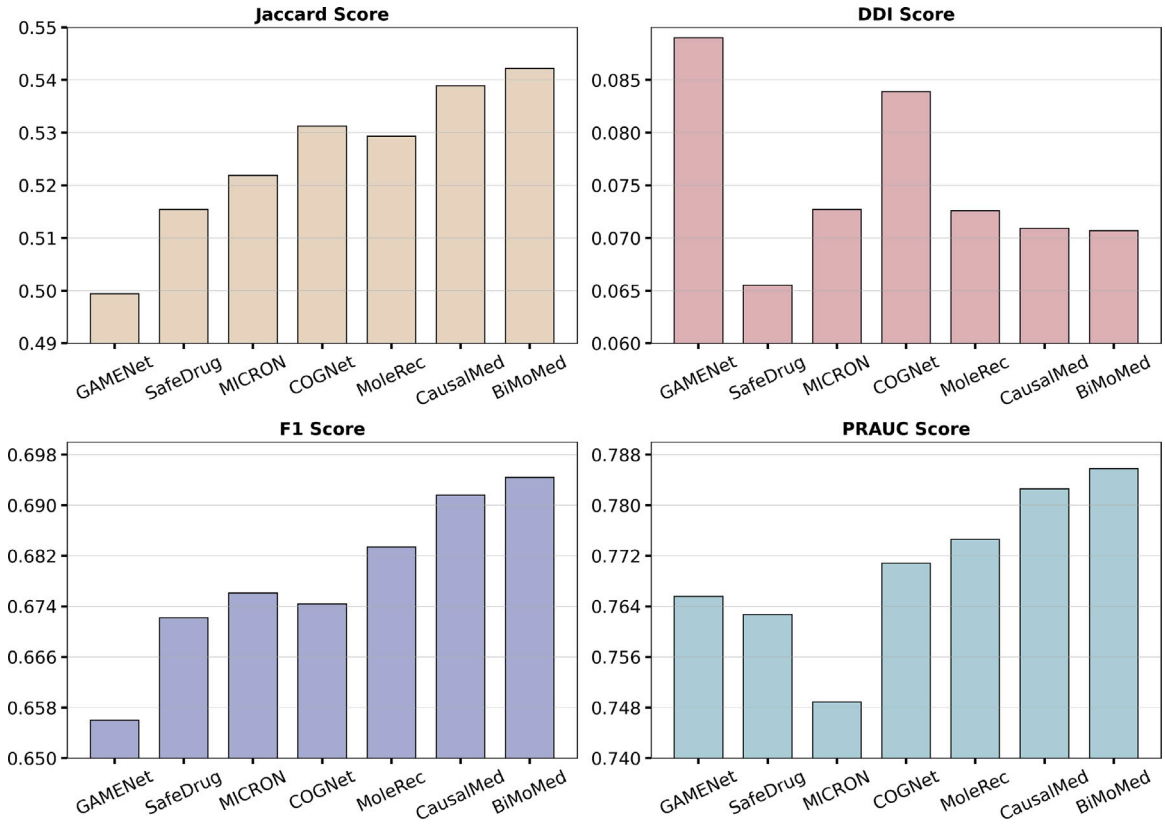


Fig. 3. Comparison with state-of-the-art baselines across all metrics in MIMIC-III.

GAMENet significantly improves model accuracy compared to previous models by incorporating patient visit history into the model framework without notably increasing complexity. However, it suffers from a high DDI rate. SafeDrug significantly reduces the DDI rate by explicitly modeling relationships between medication molecules, but it does not capture relationships between medications and diseases, thereby leaving accuracy improvement potential. MICRON adjusts recommended medication combinations by capturing changes in patients' physical conditions between two visits, resulting in improved accuracy and training efficiency after certain data filtering. COGNet designs a replication mechanism that significantly improves the Jaccard metric while increasing the number of iterations, thereby affecting time efficiency and compromising the safety of some medication combinations. MoleRec incorporates external molecular knowledge by establishing relationships between medication molecules, their substructures, and patients' diseases and procedures. This approach enhances accuracy without significantly compromising safety. However, it does not account for interactions between molecules or the effects of single molecular modalities. CausalMed recommends direct causal relationships by exploring the causal connections between medical entities, which not only improves accuracy but also enhances the interpretability of medication recommendations. Nonetheless, it does not consider relationships between entities at the molecular level.

Our model introduces the 3D modality of molecules, recalibrating the representations of molecules and their substructures, thereby enhancing molecular dimensionality and resolving structural ambiguity. We design a molecular multi-step enhancement mechanism, refining the interaction between molecular knowledge and the patient's health condition, improving the model's logic and accuracy. This approach only slightly increases the DDI rate, maintaining a balance between accuracy and safety.

In terms of operational efficiency, although the training time per epoch for our model is relatively long, the number of epochs required for convergence is greatly reduced in practice, often needing only 15 epochs. As a result, the total training time remains comparable to MoleRec (Yang et al., 2023), which also utilizes molecular knowledge.

6. Discussions

In this section, we conduct an in-depth analysis of the experimental outcomes previously discussed. Additionally, we carry out a series of supplementary experiments to confirm the comprehensiveness and rationale of our approach, to ensure its robustness and validate its advancements in the field.

6.1. Effectiveness analysis

The comparative experiments above showcase the superior performance of our model in terms of both accuracy and safety.

Previous studies on medication recommendation typically base their recommendations on the patient's physical condition. With the introduction of external molecular knowledge, both accuracy and safety are further enhanced. However, the molecular structures introduced are merely 2D projections of the actual molecular configurations, resulting in structural ambiguity. Furthermore, molecular substructures fail to independently interact with the state of a single patient visit, leading to the absence of crucial substructure combinations necessary for each visit's medication.

To address these issues, we introduce the 3D modality of molecules to supplement the missing three-dimensional information in molecular representations, thereby resolving the issue of structural ambiguity. Simultaneously, we design a molecular multi-step enhancement mechanism to optimize the use of molecular embeddings, capturing the interactions within the molecular set during a single visit and making critical molecular adjustments for each patient visit. Finally, we implement the final enhancement in response to the patient's most recent condition. Our proposed method addresses issues overlooked by other models, substantially improving the overall efficacy of the medication recommendation system.

Table 4
Complexity analysis of molecular representations in the pre-training phase: GVP vs. EGNN.

Stage	GVP	EGNN
Message generation	–	$O(G_e \cdot h^2)$
Message aggregation	–	$O(G_e \cdot h)$
Scalar feature transformation	$O(G_n \cdot s_{in} \cdot s_{out})$	$O(G_n \cdot h^2)$
Vector feature transformation	$O(G_n \cdot (s_{in} v_{out} v_{in} + v_{out} v_{in} k))$	–
Coordinate update	–	$O(G_e \cdot h^2)$

6.2. Complexity analysis

To provide a comprehensive assessment of our model's computational efficiency, we conduct a stepwise analysis of its time complexity.

Specifically, (1) In the pre-training phase, we employ GIN and GVP networks to generate representations for 2D and 3D molecular modalities, respectively. The time complexity of the GIN network is $O(L \cdot (G_n \cdot d^2 + G_e \cdot d))$, with G_n representing the number of nodes, G_e the number of edges, d the size of the embedding dimension, and L denoting the number of GIN layers. The time complexity of the GVP network is $O(G_n \cdot (s_{in} s_{out} + s_{in} v_{out} v_{in} + v_{out} v_{in} k))$, with s and k representing the dimensions of scalar and vector features, and v represents the number of channels of the vector. (2) Following this, we leverage the NTXent loss for dual-modal contrastive learning. The time complexity of the NTXent layer is $O(B^2)$, where B represents the number of molecular samples. (3) Finally, the model proceeds to the multi-step molecular enhancement phase. In this phase, we utilize embedding layers, GRU layers, molecular structure and substructure graph neural networks, and attention mechanisms for computation. The time complexity of the embedding layer is $O(L_s \cdot d)$, where L_s represents the input sequence length. The time complexity of the GRU layer is $O(g \cdot h_g^2)$, where g is the sequence length and h_g is the size of the hidden layer. The time complexity of the GNN layer is $O(L \cdot (G_n \cdot d^2 + G_e \cdot d))$, and the time complexity of the attention mechanism is $O(L_s^2 \cdot d)$.

In addition, we compare the complexity of EGNN (Satorras et al., 2021), a commonly used molecular representation method, with GVP in the pre-training phase of our study. Specifically, the molecular representation can be divided into several key steps, as shown in Table 4. The total time complexity of the EGNN network is $O((G_e + G_n) \cdot h^2)$, where the dominant term $G_e \cdot h^2$ corresponds to the message generation and coordinate update, and $G_n \cdot h^2$ corresponds to scalar feature transformation. The parameter h denotes the feature dimension of the nodes or edges. In contrast, the complexity of GVP is only linearly related to G_n, making it more suitable for extracting geometric features and encoding molecules while maintaining lower computational complexity.

Through these analyses, we found that compared to EGNN, GVP exhibits lower complexity in our model while achieving superior performance.

6.3. Ablation study

To evaluate the contribution of our innovations to the overall performance of the model and minimize interference from external factors, we remove key modules and make logical modifications to set up several variant experiments.

BiMoRec w/o **T**: In this variant, the pre-training module is removed, and the model's 3D modality data is omitted. SMILES strings are used directly during training to construct graph data for the GIN, generating embeddings for molecular structures and substructures without contrastive learning for the molecular and substructure encoders.

BiMoRec w/o **E**: In this variant, we removed the molecular multi-step enhancement module, thereby eliminating the interaction between molecules and single patient visits, along with intermolecular and latest visit enhancements. Instead, we used aggregated patient representations derived from comprehensive patient histories to identify the required molecules, which were then transformed into probabilities for generating medication recommendations.

BiMoRec w/o **T+E**: In this variant, we concurrently remove both the pre-training module and the molecular multi-step enhancement module. This variant does not integrate the dual modalities of molecular representation, nor does it interact with the patient's single representation, relying solely on the relationships between medications, diseases, and procedures in patient visits to make recommendations.

As shown in Table 5, the ablation experiment results align with our expectations.

BiMoRec w/o T demonstrates that by introducing the 3D molecular modality, it improves recommendation accuracy, supplements the missing three-dimensional spatial information of the molecules, resolves structural ambiguity, and optimizes the molecular representation space, as further substantiated in the subsequent case analysis.

BiMoRec w/o E shows that our multi-step enhancement module captures relevant molecular substructures during each visit, thereby enhancing the model's performance. This demonstrates that capturing the relationship between patient representations and molecular structures at visit-level granularity is advantageous for model performance.

BiMoRec w/o T+E exhibits a slight improvement in safety but a noticeable decrease in accuracy. This phenomenon is common in medication recommendation systems, as there is often a necessary balance between safety and accuracy, with the best possible performance typically resulting from a trade-off between the two. Furthermore, the experiment demonstrates that the molecular bimodal mechanism and the multi-step enhancement mechanism, along with patient representation interactions, exhibit a synergistic effect, and their combined use further enhances model performance.

Furthermore, to better evaluate the effectiveness of our model in 3D molecular representation tasks, we compare the GVP used in our study with two other classical molecular representation models: EGNN (Satorras et al., 2021) and SE(3)-Transformer (Fuchs et al., 2020). Specifically, EGNN maintains equivariance through message passing and coordinate updates, making it well-suited for capturing molecular dynamic geometry. And SE(3)-Transformer improves upon EGNN by explicitly incorporating vector representations into its processing, ensuring equivariance to rotations and translations. This enhancement helps address some of EGNN's limitations and generally leads to more expressive and accurate molecular representations.

The experimental results, as shown in Table 6, demonstrate that GVP consistently outperforms both models across various metrics, with SE(3)-Transformer ranking second and EGNN performing the worst. This disparity may stem from EGNN's lack of explicit separation between scalar and vector features, which limits its ability to effectively capture directional information, ultimately leading to inferior performance. While SE(3)-Transformer enhances EGNN by incorporating vector representations, its complex attention mechanism can introduce unstable high-order geometric noise, making the contrastive objective more challenging to converge. Furthermore, its increased model complexity and higher data requirements may hinder its effectiveness in medication recommendation tasks, where more specialized networks tend to perform better. As a result, SE(3)-Transformer yields suboptimal final results compared to GVP.

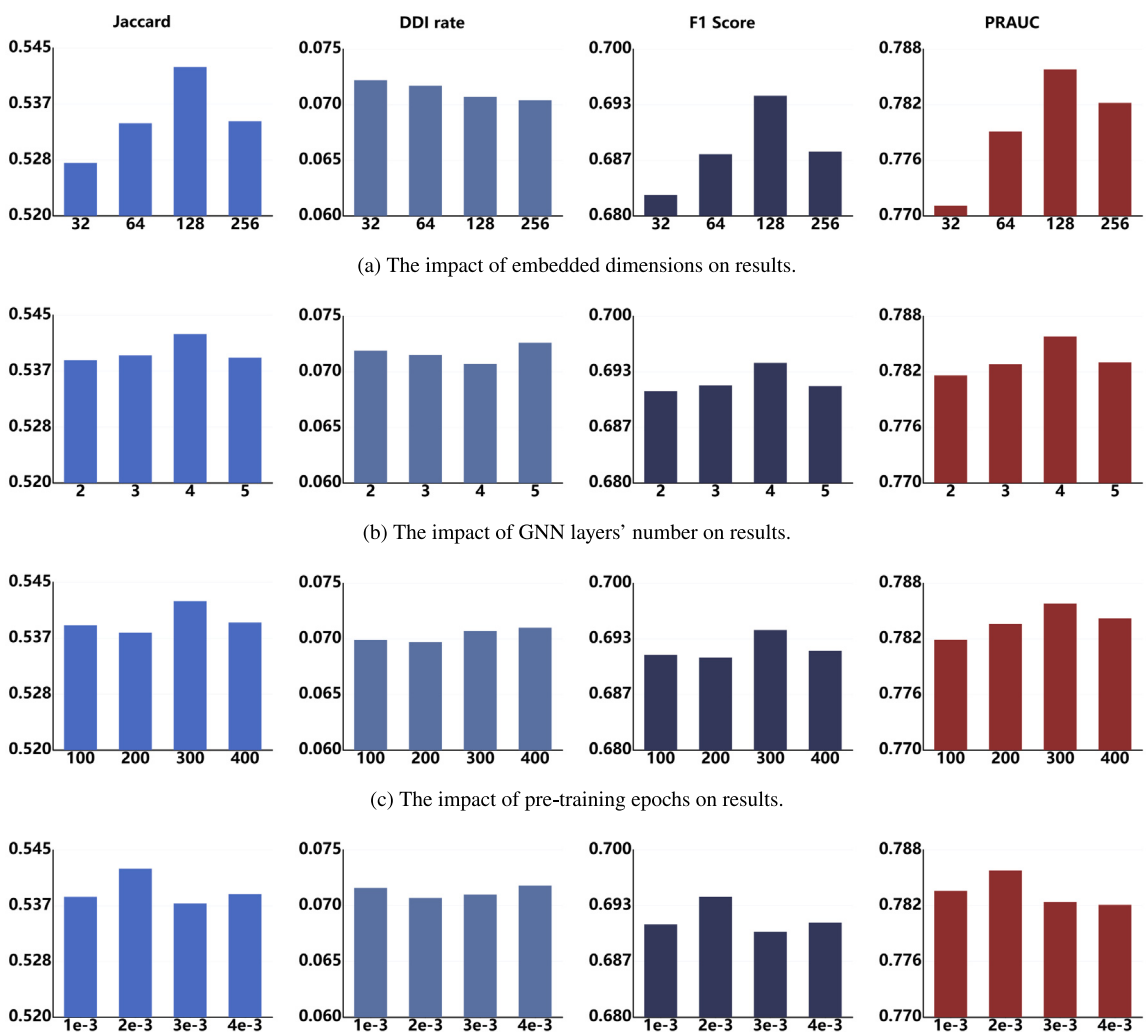
In contrast, GVP explicitly distinguishes vector features, enabling it to extract geometric information more effectively in molecular processing. This advantage makes it particularly well-suited for the molecular contrastive learning stage in BiMoRec.

Table 5
The performance of each ablation model on the test set regarding accuracy and safety. The best and the runner-up results are highlighted in bold and underlined respectively.

Model	MIMIC-III					MIMIC-IV				
	Jaccard↑	DDI↓	F1↑	PRAUC↑	Avg.#Med	Jaccard↑	DDI↓	F1↑	PRAUC↑	Avg.#Med
BiMoRec w/o T	0.5356	**0.0697**	0.6887	0.7823	20.4086	0.4832	0.0704	0.6348	0.7268	13.5401
BiMoRec w/o E	0.5347	0.0735	0.6882	0.7752	21.8691	0.4800	0.0698	0.6320	0.7231	15.5599
BiMoRec w/o T+E	0.5335	0.0747	0.6873	0.7782	22.7640	0.4778	0.0696	0.6301	0.7212	15.9020
BiMoRec	**0.5422**	0.0707	**0.6944**	**0.7858**	20.3989	**0.4917**	**0.0696**	**0.6425**	**0.7366**	13.5291

Table 6
Comparison of experimental results from different molecular 3D encoders. The best and the runner-up results are highlighted in bold and underlined respectively.

Model	MIMIC-III					MIMIC-IV				
	Jaccard↑	DDI↓	F1↑	PRAUC↑	Avg.#Med	Jaccard↑	DDI↓	F1↑	PRAUC↑	Avg.#Med
BiMoRec$_{\text{EGNN}}$	0.5363	0.0708	0.6898	0.7819	19.9768	0.4795	0.0703	0.6312	0.7239	13.7881
BiMoRec$_{\text{SE(3)-Transformer}}$	0.5395	0.0722	0.6925	0.7836	20.7264	0.4909	**0.0683**	0.6417	0.7360	13.3506
BiMoRec	**0.5422**	**0.0707**	**0.6944**	**0.7858**	20.3989	**0.4917**	0.0696	**0.6425**	**0.7366**	13.5291



(a) The impact of embedded dimensions on results.

(b) The impact of GNN layers' number on results.

(c) The impact of pre-training epochs on results.

(d) The impact of pre-training learning rate on results.

Fig. 4. Experiments on parameter sensitivity on MIMIC-III dataset.

6.4. Parameter sensitivity

To investigate the influence of specific parameters on the model, we conduct a series of parameter sensitivity experiments on the MIMIC-III dataset, focusing on four key model parameters. These experiments examine the effects of different parameter combinations on model performance and analyze the underlying causes of these changes in detail. The results are depicted in Fig. 4, where the X-axis represents different parameter values, and the Y-axis shows the performance variations caused by these values.

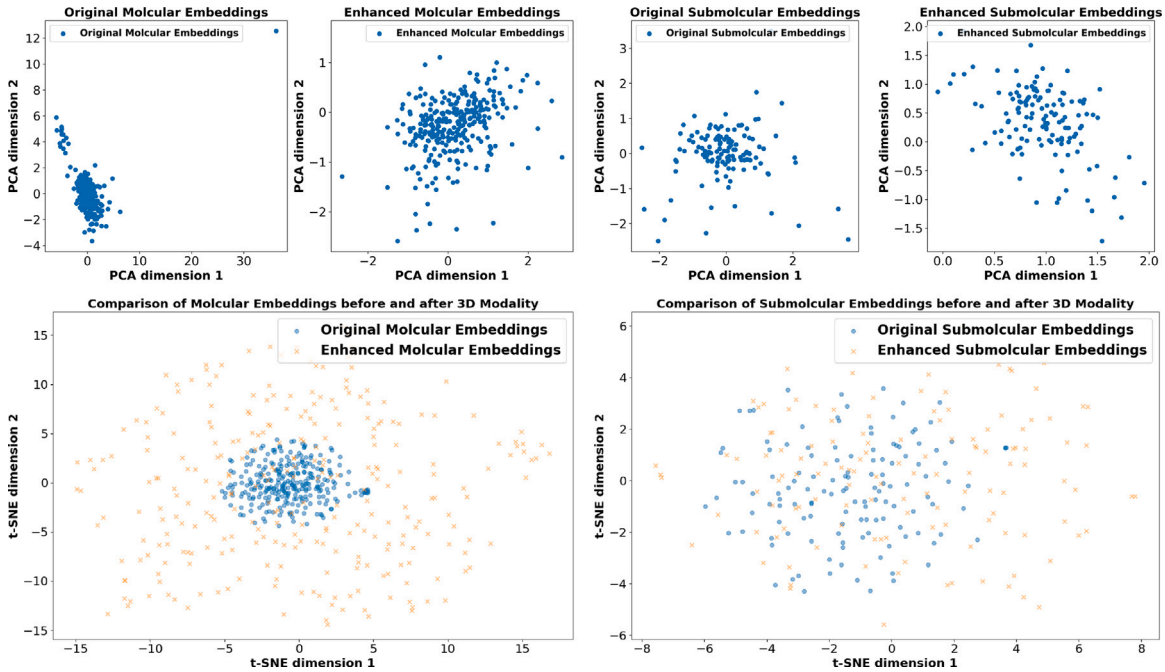


Fig. 5. Low dimensional data distribution represented by molecular and substructure embeddings.

First, we examine the effect of different embedding dimensions on model performance. Fig. 4(a) illustrates performance at embedding dimensions of 32, 64, 128, and 256. When the embedding size is 128, the model achieves significantly higher accuracy, while safety remains consistent. Therefore, considering accuracy, safety, and computational cost, we select 128 as the optimal embedding dimension for this experiment.

Next, since both molecular modalities use GNN layers as the basic encoder, we assess the impact of the number of GNN layers. Fig. 4(b) shows that a 4-layer network provides superior accuracy. However, too many layers increase model complexity and the likelihood of overfitting, leading to a larger gap between training and testing results. Conversely, fewer layers reduce convergence speed, preventing the model from achieving optimal performance. Thus, we select 4 layers for the model.

Regarding the number of epochs and learning rate of the pre-training model, these are critical to the performance of the pre-trained model. As shown in Figs. 4(c) and 4(d), too few epochs do not sufficiently perform contrastive learning for the dual molecular modalities, while excessive epochs offer diminishing returns and may reduce performance. The pre-training learning rate also requires fine-tuning, and an optimizer strategy for adjusting the learning step size is recommended.

In practical model experiments, parameter adjustments may be required based on the experimental environment and actual results. Different experimental environments may lead to minor variations.

6.5. Case study

To further clarify the reasoning behind our innovations, we conduct a case study using a sample from MIMIC-III. This specific instance highlights the presence of structural confusion and how it interferes with molecular representation and medication recommendation. It also emphasizes the importance of the interaction between molecular knowledge and individual patient visit representations in identifying key substructures for each visit.

For example, the molecular structures of Atropine and Ipratropium are similar, yet exhibit distinct functional differences. Atropine is primarily used to treat conditions such as blurred vision and arrhythmia, while Ipratropium is primarily used to treat COPD (chronic obstructive pulmonary disease) and other respiratory diseases, administered via nebulization. As shown in Fig. 6, by incorporating the 3D modality, we successfully distinguish between medications such as R01A and A03B, N07A, and recommend the correct medication to the user. As shown in Fig. 5, we analyze the low-dimensional representation of molecular embeddings generated by both the original molecular encoder and the newly introduced bimodal encoder. The more uniform distribution in the PCA (Principal Component Analysis) (Jolliffe, 2002) plot indicates that significant variability is captured, facilitating molecular differentiation. Similarly, the cluster distribution in the t-SNE (t-distributed Stochastic Neighbor Embedding) (Van der Maaten & Hinton, 2008) plot further supports this observation.

In summary, the introduction of 3D modality data addresses structural confusion and effectively optimizes the entire molecular representation space. The subsequent multi-step molecular enhancement mechanism ensures that the optimized representations engage with patient representations at a finer granularity, thereby selecting the medication molecules that the patient truly needs. Case studies validate the above analysis and confirm the effectiveness of our approach.

Table 7
The performance of model migration on SafeDrug and MoleRec.

Method	Jaccard	DDI	Training time /Epoch (s)
SafeDrug	0.5093	0.0632	256.15
MoleRec	0.5307	0.0745	402.92
BiMoRec	0.5422	0.0707	416.61

6.6. Model scalability

We transfer the contrastive learning module of our model to the SafeDrug and MoleRec models. Table 7 shows the corresponding model performance and training efficiency. It is important to note that during the transfer, we standardized the hyperparameters, which influenced certain metrics. Nevertheless, the results clearly demonstrate the high scalability of our model, highlighting its ease of transfer to existing molecular-related frameworks.

	Molecular		Patient 432			Patient 389		
	Atropine	Ipratropium	BiMoRec	BiMoRec_T+E	True results	BiMoRec	BiMoRec_T+E	True results
R01A	√	×	81.73%	31.47%	√	8.31%	94.32%	×
N07A	×	√	39.40%	93.68%	×	100%	98.93%	√
A03B	×	√	4.04%	2.50%	×	64.19%	20.16%	√

Fig. 6. Detailed case studies on molecular 3D modes and multi enhancement in MIMIC-III dataset.

7. Conclusion

This paper presents the BiMoRec medication recommendation model, which addresses the challenge of structural ambiguity through the integration of dual molecular modalities. This approach greatly improves molecular differentiation, optimizing the representation space of molecules. Furthermore, by employing a designed molecular multi-step enhancement mechanism, the interactions between molecules and the relationships between patients and molecules are refined at the visit level. In the final stage, the model optimizes the representation based on the patient's most recent visit, identifying truly effective medication molecules, thereby significantly improving the accuracy and safety of the model. We conducted a series of rigorous experiments on publicly available clinical datasets, validating the effectiveness of our approach.

This study introduces a novel molecular modality and recommendation mechanism, highlighting related challenges and which may provide a foundation for future research. Our future goal is to continue advancing medication recommendation research by exploring large models and their interactions between multiple modalities to optimize our framework, thereby addressing more complex problems and real-world scenarios.

CRediT authorship contribution statement

Shi Mu: Conceptualization, Methodology, Software, Validation, Formal analysis, Data Curation, Investigation, Writing – original draft, Writing – review & editing, Visualization. **Chen Li:** Writing – original draft, Writing – review & editing, Supervision. **Xiang Li:** Writing – review & editing. **Shunpan Liang:** Supervision, Funding acquisition.

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Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Shunpan Liang reports financial support was provided by Hebei Provincial Natural Science Foundation and Xinjiang Uygur Autonomous Region Natural Science Foundation. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

An, Y., Zhang, L., You, M., Tian, X., Jin, B., & Wei, X. (2021). MeSIN: Multilevel selective and interactive network for medication recommendation. *Knowledge-Based Systems*, *233*, Article 107534.

Bhoi, S., Lee, M. L., Hsu, W., Fang, H. S. A., & Tan, N. C. (2021). Personalizing medication recommendation with a graph-based approach. *ACM Transactions on Information Systems (TOIS)*, *40*(3), 1–23.

Chen, T., Kornblith, S., Norouzi, M., & Hinton, G. (2020). A simple framework for contrastive learning of visual representations. *Cornell University - ArXiv*, Cornell University - ArXiv.

Chen, Q., Li, X., Geng, K., & Wang, M. (2023). Context-aware safe medication recommendations with molecular graph and DDI graph embedding. *vol. 37*, In *Proceedings of the AAAI conference on artificial intelligence* (pp. 7053–7060).

Choi, E., Bahadori, M. T., Sun, J., Kulas, J., Schuetz, A., & Stewart, W. (2016). Retain: An interpretable predictive model for healthcare using reverse time attention mechanism. *Advances in Neural Information Processing Systems*, *29*.

Dauji, S., & Bhargava, K. (2016). Estimation of concrete characteristic strength from limited data by bootstrap. *Journal of Asian Concrete Federation*, *2*(1), 81–94.

Fuchs, F., Worrall, D., Fischer, V., & Welling, M. (2020). Se (3)-transformers: 3d roto-translation equivariant attention networks. *Advances in Neural Information Processing Systems*, *33*, 1970–1981.

Han, J., Rong, Y., Xu, T., & Huang, W. (2022). Geometrically equivariant graph neural networks: A survey. arXiv preprint arXiv:2202.07230.

Hoffman, K. B., Dimbil, M., Erdman, C. B., Tatonetti, N. P., & Overstreet, B. M. (2014). The Weber effect and the United States food and drug administration's adverse event reporting system (FAERS): analysis of sixty-two drugs approved from 2006 to 2010. *Drug Safety*, *37*, 283–294.

Isert, C., Atz, K., & Schneider, G. (2023). Structure-based drug design with geometric deep learning. *Current Opinion in Structural Biology*, *79*, Article 102548.

Jiang, S., Feng, F., Chen, W., Li, X., & He, X. (2021). Structure-enhanced meta-learning for few-shot graph classification. *AI Open*, *2*, 160–167.

Jin, B., Yang, H., Sun, L., Liu, C., Qu, Y., & Tong, J. (2018). A treatment engine by predicting next-period prescriptions. In *Proceedings of the 24th ACM SIGKDD international conference on knowledge discovery & data mining* (pp. 1608–1616).

Jing, B., Eismann, S., Suriana, P., Townshend, R., & Dror, R. (2020). Learning from protein structure with geometric vector perceptrons. *Learning*, Learning.

Johnson, A. E., Bulgarelli, L., Shen, L., Gayles, A., Shammout, A., Horng, S., et al. (2023). MIMIC-IV, a freely accessible electronic health record dataset. *Scientific Data*, *10*(1), 1.

Johnson, A. E., Pollard, T. J., Shen, L., Lehman, L.-w. H., Feng, M., Ghassemi, M., et al. (2016). MIMIC-III, a freely accessible critical care database. *Scientific Data*, *3*(1), 1–9.

Jolliffe, I. T. (2002). *Principal Component Analysis for Special Types of Data*. Springer.

Kim, S., Chen, J., Cheng, T., Gindulyte, A., He, J., He, S., et al. (2021). PubChem in 2021: new data content and improved web interfaces. *Nucleic Acids Research*, *49*(D1), D1388–D1395.

Landrum, G. (2013). Rdkit documentation. *Release*, *1*(1–79), 4.

Le, H., Tran, T., & Venkatesh, S. (2018). Dual memory neural computer for asynchronous two-view sequential learning. In *Proceedings of the 24th ACM SIGKDD international conference on knowledge discovery & data mining* (pp. 1637–1645).

Lee, J., Lee, Y., Kim, J., Kosiorek, A., Choi, S., & Teh, Y. (2019). Set transformer: A framework for attention-based permutation-invariant neural networks. *International Conference on Machine Learning*, International Conference on Machine Learning.

Li, X., Liang, S., Hou, Y., & Ma, T. (2023). StratMed: Relevance stratification between biomedical entities for sparsity on medication recommendation. *Knowledge-Based Systems*, Article 111239.

Li, X., Liang, S., Lei, Y., Li, C., Hou, Y., Zheng, D., et al. (2024). CausalMed: Causality-based personalized medication recommendation centered on patient health state. In *Proceedings of the 33rd ACM international conference on information and knowledge management* (pp. 1276–1285).

Li, S., Zhou, J., Xu, T., Dou, D., & Xiong, H. (2022). Geomgcl: Geometric graph contrastive learning for molecular property prediction. *vol. 36*, In *Proceedings of the AAAI conference on artificial intelligence* (pp. 4541–4549).

Liu, S., Wang, H., Liu, W., Lasenby, J., Guo, H., & Tang, J. (2021). Pre-training molecular graph representation with 3d geometry. arXiv preprint arXiv:2110.07728.

Luo, Y., Liu, Y., & Peng, J. (2023). Calibrated geometric deep learning improves kinase–drug binding predictions. *Nature Machine Intelligence*, *5*(12), 1390–1401.

Van der Maaten, L., & Hinton, G. (2008). Visualizing data using t-SNE. *Journal of Machine Learning Research*, *9*(11).

Meng, X.-Y., Zhang, H.-X., Mezei, M., & Cui, M. (2011). Molecular docking: a powerful approach for structure-based drug discovery. *Current Computer-Aided Drug Design*, *7*(2), 146–157.

Miao, S., Liu, M., & Li, P. (2022). Interpretable and generalizable graph learning via stochastic attention mechanism. In *International conference on machine learning* (pp. 15524–15543). PMLR.

Mishra, R., & Shridevi, S. (2024). Knowledge graph driven medicine recommendation system using graph neural networks on longitudinal medical records. *Scientific Reports*, *14*(1), 25449.

Picheny, V., Kim, N. H., & Haftka, R. T. (2010). Application of bootstrap method in conservative estimation of reliability with limited samples. *Structural and Multidisciplinary Optimization*, *41*, 205–217.

Read, J., Pfahringer, B., Holmes, G., & Frank, E. (2011). Classifier chains for multi-label classification. *Machine Learning*, *85*, 333–359.

Satorras, V. G., Hoogeboom, E., & Welling, M. (2021). E (n) equivariant graph neural networks. In *International conference on machine learning* (pp. 9323–9332). PMLR.

Shang, J., Xiao, C., Ma, T., Li, H., & Sun, J. (2019). Gamenet: Graph augmented memory networks for recommending medication combination. *vol. 33*, In *Proceedings of the AAAI conference on artificial intelligence* (pp. 1126–1133).

Stark, H., Beaini, D., Corso, G., Tossou, P., Dallago, C., Günnemann, S., et al. (2021). 3D infomax improves GNNs for molecular property prediction. ArXiv: Learning, ArXiv: Learning.

Townshend, R. J., Vögele, M., Suriana, P., Derry, A., Powers, A., Laloudakis, Y., et al. (2020). Atom3d: Tasks on molecules in three dimensions. arXiv preprint arXiv:2012.04035.

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A., et al. (2017). Attention is all you need. *Neural Information Processing Systems*, Neural Information Processing Systems.

Verma, J., Khedkar, V. M., & Coutinho, E. C. (2010). 3D-QSAR in drug design-a review. *Current Topics in Medicinal Chemistry*, *10*(1), 95–115.

Wang, Z., Liang, Y., & Liu, Z. (2022). FFBDNet: Feature fusion and bipartite decision networks for recommending medication combination. In *Joint European conference on machine learning and knowledge discovery in databases* (pp. 419–436). Springer.

Wang, M., Liu, M., Liu, J., Wang, S., Long, G., & Qian, B. (2017). Safe medicine recommendation via medical knowledge graph embedding. arxiv. *Information Retrieval*.

Wang, Y., Wang, J., Cao, Z., & Barati Farimani, A. (2022). Molecular contrastive learning of representations via graph neural networks. *Nature Machine Intelligence*, *4*(3), 279–287.

Wang, Y., Zhang, J., Guo, S., Yin, H., Li, C., & Chen, H. (2021). Decoupling representation learning and classification for gnn-based anomaly detection. In *Proceedings of the 44th international ACM SIGIR conference on research and development in information retrieval* (pp. 1239–1248).

Wu, J., He, K., Mao, R., Li, C., & Cambria, E. (2023). MEGACare: Knowledge-guided multi-view hypergraph predictive framework for healthcare. *Information Fusion*, *100*, Article 101939.

Wu, R., Qiu, Z., Jiang, J., Qi, G., & Wu, X. (2022). Conditional generation net for medication recommendation. In *Proceedings of the ACM web conference 2022* (pp. 935–945).

Wu, F., Wu, L., Radev, D., Xu, J., & Li, S. Z. (2023). Integration of pre-trained protein language models into geometric deep learning networks. *Communications Biology*, *6*(1), 876.

Wu, J., Yu, X., He, K., Gao, Z., & Gong, T. (2024). PROMISE: A pre-trained knowledge-infused multimodal representation learning framework for medication recommendation. *Information Processing & Management*, *61*(4), Article 103758.

Xu, K., Hu, W., Leskovec, J., & Jegelka, S. (2018). How powerful are graph neural networks?. arXiv preprint arXiv:1810.00826.

Yang, C., Xiao, C., Glass, L., & Sun, J. (2021). Change matters: Medication change prediction with recurrent residual networks. In *Proceedings of the thirtieth international joint conference on artificial intelligence 2021* (pp. 3728–3734).

Yang, C., Xiao, C., Ma, F., Glass, L., & Sun, J. (2021). SafeDrug: Dual molecular graph encoders for safe drug recommendations. In *Proceedings of the thirtieth international joint conference on artificial intelligence* (pp. 3735–3741).

Yang, N., Zeng, K., Wu, Q., & Yan, J. (2023). Molerec: Combinatorial drug recommendation with substructure-aware molecular representation learning. In *Proceedings of the ACM web conference 2023* (pp. 4075–4085).

Yu, J.-L., Wang, Y.-G., Peng, J., Wu, J.-W., Zhou, C., & Li, G.-B. (2024). Geometric deep learning-enabled metal-binding site identification and grafting. *Fundamental Research*.

Zhang, Y., Chen, R., Tang, J., Stewart, W. F., & Sun, J. (2017). LEAP: learning to prescribe effective and safe treatment combinations for multimorbidity. In *Proceedings of the 23rd ACM SIGKDD international conference on knowledge discovery and data mining* (pp. 1315–1324).

Zhang, K., Zhang, Z., Wang, W., Liang, Y., & Wang, X. (2024). Hyper-relational knowledge enhanced network for hypertension medication recommendation. *IEEE Transactions on Computational Social Systems*.

Zheng, Z., Wang, C., Xu, T., Shen, D., Qin, P., Huai, B., et al. (2021). Drug package recommendation via interaction-aware graph induction. In *Proceedings of the web conference 2021* (pp. 1284–1295).

Zheng, Z., Wang, C., Xu, T., Shen, D., Qin, P., Zhao, X., et al. (2023). Interaction-aware drug package recommendation via policy gradient. *ACM Transactions on Information Systems*, *41*(1), 1–32.

Zhu, J., Xia, Y., Wu, L., Xie, S., Qin, T., Zhou, W., et al. (2022). Unified 2d and 3d pre-training of molecular representations. In *Proceedings of the 28th ACM SIGKDD conference on knowledge discovery and data mining* (pp. 2626–2636).